Lecture 23: Population Health Data & Survival Analysis

BINF 4002 — Machine Learning for Health

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Where We Are

Modality progression:

- L17–18: EHR & Claims
- L19–20: Clinical Text & NLP
- L21–22: Clinical Imaging

→ L23: Population Health Data & Survival Analysis ← **today**

- ○ L24: Causality & Fairness
- L25+: Omics, Molecules, Deployment

Today has two parts:

1. **Population health data:** registries, biobanks, and clinical trials—*designed* data, in contrast to the *byproduct* data of L17–22.
2. **Survival analysis:** a modeling technique for time-to-event outcomes, common in this domain and many others.

Part I: Population Health Data

A Question

Which treatment for kidney stones is better—open surgery (A) or percutaneous nephrolithotomy (B)?

Attempt 1: Use the EHR.

You have your hospital's data. Some patients received Treatment A, some B. You could compare their success rates.

What could go wrong?

Why the EHR Isn't Enough

Several problems:

- **One health system.** Specific patient population, geographic bias—not representative.

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So EHR data alone won't give us a reliable answer. What else could we try?

Attempt 2: Registry Data

Suppose we survey many hospitals and individuals. Collect data on everyone who had Treatment A or B, their outcomes, and key patient characteristics. This is **registry data**—designed to capture a broad population.

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The result:

Treatment A (open surgery): 78% success.   Treatment B (PCNL): 83% success.

Treatment B looks better. Done?

Not So Fast: Simpson's Paradox

Kidney stone treatment data (Charig et al., 1986):

	Open surgery (A)	PCNL (B)	"Better"
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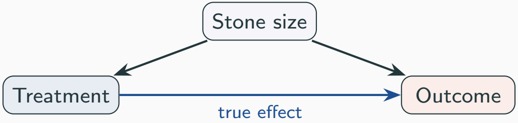
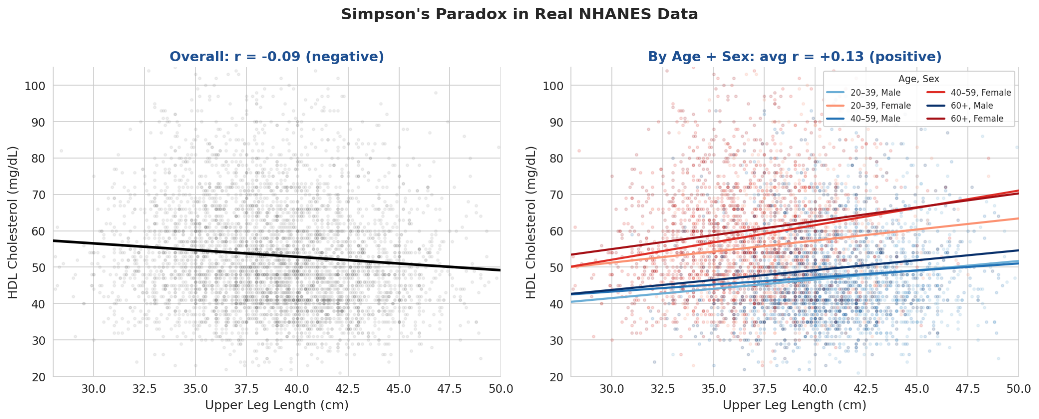
Large stones	73%	69%	A

A is better in *every subgroup*, but worse overall. Open surgery was assigned to large (harder) stones—confounding by severity.



Simpson’s Paradox in Real Health Data



Real NHANES 2017–2018 data. Overall, longer legs correlate with *lower* HDL cholesterol ($r = -0.09$). Within every age+sex subgroup, the relationship reverses (avg $r = +0.13$). Men have longer legs *and* lower HDL—aggregating across sex and age reverses the within-group trend.

What Registries Can and Can't Tell You

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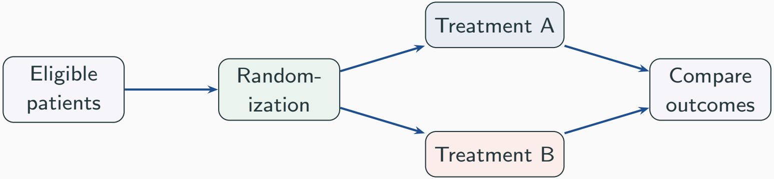
Registries struggle with:

- Answering *what would have happened under a different treatment*
- Confounders must be measured *and* adjusted for—Simpson's Paradox shows what goes wrong when they aren't

For the most reliable causal answer, we'd need randomization.

Randomized Trials



Randomization balances confounders—measured *and* unmeasured—in expectation.

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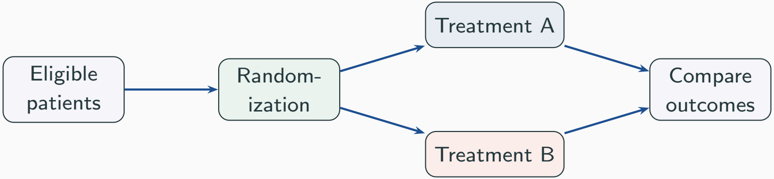

Randomization balances confounders—measured *and* unmeasured—in expectation.

But: expensive (\$10K–50K per patient for Phase III), sometimes unethical, and the enrolled population is narrow.

How narrow? Murthy et al. (2004): $<$25% of cancer trial participants were $>$65, despite $>$60% of cancer in that age group. Travers et al. (2007): only $\sim$4% of community asthma patients would have met eligibility for the major RCTs behind their treatment guidelines.

The Question Determines the Data

Question type	Example	Data source	Key risk
Descriptive	What is the 5-year survival rate for this cancer?	Registry, cohort	Selection bias
Predictive	Will *this patient* respond to treatment?	Any with outcomes	Distribution shift
Causal	Does Treatment A *cause* better outcomes than B?	RCT; or observational with causal methods	Confounding

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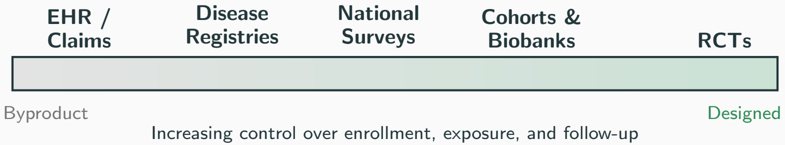
Causal inference methods (propensity scores, instrumental variables, etc.) can enable causal claims from observational data—including EHR—if done carefully. Example: COVID-19 vaccine effectiveness studies from EHR data. **Thursday (L24)** formalizes this.

The Data Spectrum



These are *alternative* data sources, not stages. Modern biobanks blur the boundary.

Next: we'll look at specific registries, biobanks, and trial designs in more depth.

Data Sources in Depth

Registries & Cohorts

What Does Designed Population Data Look Like?

NHANES (National Health and Nutrition Examination Survey) is a good example of designed population data:

- About 9,000 examined participants per 2-year cycle
- Home interview (demographics, medical history, diet) + clinic visit (physical exam, labs, questionnaires)
- Complex survey design: deliberately oversamples minorities, elderly, low-income
- Public data, no credentialing: `wwwn.cdc.gov/nchs/nhanes/`

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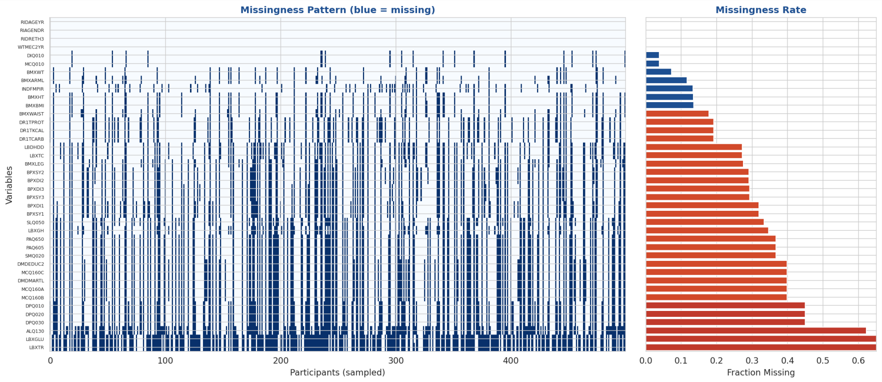
Key for ML: Unweighted statistics describe the *sample*; weighted statistics describe the *population*. Missingness is mostly by protocol ("this module wasn't administered this cycle"), not informative—unlike EHR data.

Questionnaire PDFs: `wwwn.cdc.gov/nchs/nhanes/.../questionnaires.aspx`

NHANES: Variable Structure (Notebook Fig. 1)



Real NHANES 2017–2018 data (about 9,000 participants, about 40 variables, 16 survey components). Left: missingness heatmap—note block structure. Right: per-variable fraction missing. Variable names follow CDC conventions: **BMXBMI** = Body Mass Index, **BPXSY1** = Systolic Blood Pressure (1st reading), **LBXTR** = Triglycerides, **LBXGLU** = Fasting Glucose, **DR1TKCAL** = Dietary Recall Calories, **DPQ010** = Depression Screener (PHQ-9 Q1).

SEER: Cancer Surveillance

Surveillance, Epidemiology, and End Results:

- Covers about 46% of the US population
- Cancer incidence, treatment, survival by site, stage, demographics
- Research Data: signed agreement; Research Plus: eRA Commons
- `seer.cancer.gov/data/`

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One lesson from SEER: large-scale descriptive survival data, but limited confounder control. Treatment assignment is not randomized, so hazard ratios are associational.

Limitations: No detailed treatment data, no imaging, limited labs. Geographic selection.

Biobanks: Designed Cohorts That Blur the Boundary

Biobank	N	Country	Key features
UK Biobank	$\sim$500K	UK	Imaging + genomics + GP/EHR linkage
All of Us	$\sim$850K	US	Diversity emphasis, EHR-linked, wearables
Japan Biobank	$\sim$270K	Japan	Disease-focused recruitment

One lesson from biobanks: designed enrollment with EHR linkage blurs the byproduct/designed boundary. Participants consented to research, but their clinical data is byproduct.

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ukbiobank.ac.uk allofus.nih.gov biobankjp.org

The Healthy Volunteer Effect

People who volunteer for research are systematically different.

- Fry et al. (2017): UK Biobank participants had **40–50% lower all-cause mortality** than the UK average. Response rate: 5.5%.
- Batty et al. (2020): despite strong selection into UK Biobank, many risk factor–mortality associations were similar to those in more representative cohorts—but absolute rates and participant composition are biased.
- Volunteer-based cohort studies (Framingham, etc.) generally enroll healthier, more health-conscious participants than the general population.

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All of Us deliberately oversamples underrepresented groups—but still faces within-stratum volunteer bias.

A model trained on biobank volunteers may not generalize. The bias is in *who enrolls*, not in the measurements.

Clinical Trials

RCTs Are Not One Thing

Design	What's randomized	Example	When to use
Parallel-group	Patients $\rightarrow$ arms	Most drug trials	Default design
Crossover	Patients $\rightarrow$ sequence	Chronic disease	Patient is own control
Cluster	Sites / clinics	Implementation science	Group-level intervention
Pragmatic	Patients, broad elig.	Real-world effectiveness	Generalizability
Micro-randomized	Within-person, repeated	mHealth notifications	Digital health

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ClinicalTrials.gov: ~580K registered studies. Public REST API: `clinicaltrials.gov/api`

In practice, evidence is often synthesized across studies via **meta-analysis** and **systematic reviews**—no single study is definitive.

Part II: Survival Analysis

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Registries and trials often produce **time-to-event outcomes**: time to death, recurrence, hospital readmission. Follow-up is frequently incomplete—patients drop out, studies end, competing events occur.

Standard binary classification ("did the patient die: yes/no") discards timing and mishandles incomplete follow-up.

Survival analysis is a modeling framework for these outcomes—common in oncology, cardiovascular research, and epidemiology, but useful whenever you model *when* something happens, not just *whether*.

Why Not Just Classification or Regression?

Approach 1: Binary classification—"dead at 5 years: yes/no."

- Throws away timing (a death at month 2 $\neq$ death at month 58)
- Can't use patients with $<$5 years of follow-up—must exclude them entirely

Approach 2: Just drop the censored patients and analyze the rest.

- But *why* were they censored? Did they leave the health system, move away, or just stay healthy? If censoring is related to the outcome, dropping them biases everything.

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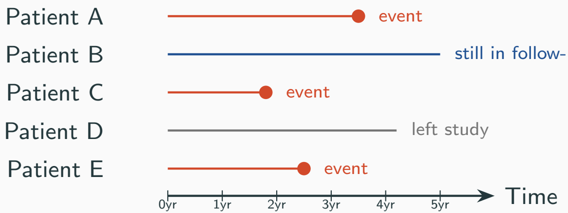
What survival analysis does differently:

- The label is a *pair* (T, δ): observed time + event indicator
- The loss function uses censored patients' information ("survived at least this long") without pretending they had the event
- You *can* use standard classifiers (predict event in each time bin)—but you must design the problem and loss carefully. This is the **discrete-time survival** approach.

Censoring and Competing Risks

Some patients haven't had the event yet. What do you do with them?



B and D are **right-censored**: survived at least this long, true event time unknown. Dropping them wastes data; treating them as events is wrong.

Key assumption: Censoring is *non-informative*—patients who drop out aren't systematically sicker or healthier. If they are, estimates are biased.

Competing risks: Patient D might have died of something unrelated—making the event of interest impossible, not just unobserved.

Survival Concepts — Before Any Model

Two quantities we want to estimate:

Survival function: $S(t) = P(T > t)$

The probability that a patient drawn from this population survives past time t. For an individual, we interpret it probabilistically: "given what we know, what's the chance this patient is still event-free at time t?" Starts at 1, decreases.

Hazard function: The instantaneous event rate among those who haven't had the event yet—"given that you've survived this long, how quickly are events happening *right now*?"

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These are mathematically linked: knowing one determines the other. **Kaplan-Meier** estimates $S(t)$. **Cox PH** models $h(t)$ as a function of covariates.

Kaplan-Meier Estimator

The intuition: At any time t, count how many patients are still being followed who haven't had the event. Each time someone has the event, the estimated survival probability steps down. Censored patients leave the count quietly—without causing a step.

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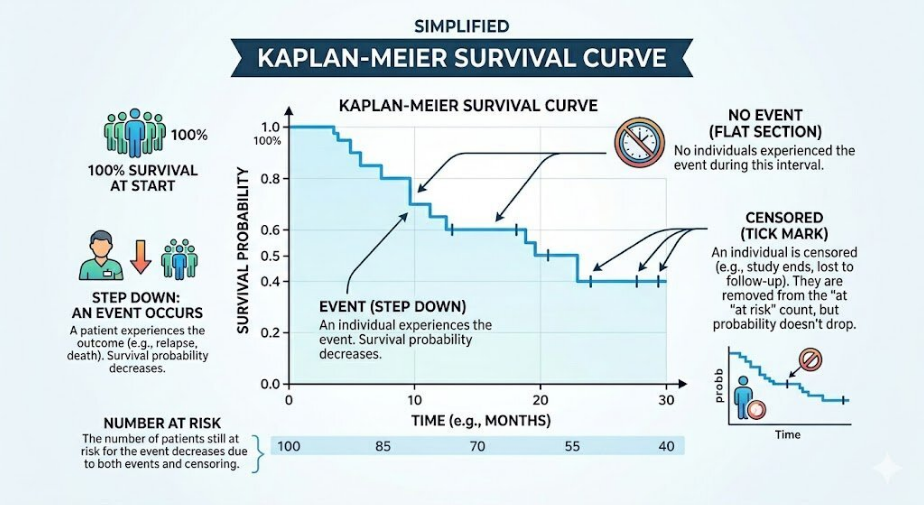
Formally: At each event time t_j,

$$\hat{S}(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{n_j}\right)$$

where d_j = events at t_j, n_j = patients still at risk (haven't had the event and haven't been censored yet).

$\hat{S}(t)$ is a **step function**: it drops at each event time and stays flat in between. **Log-rank test:** compares two K-M curves (H_0: no difference).

Reading a Kaplan-Meier Curve



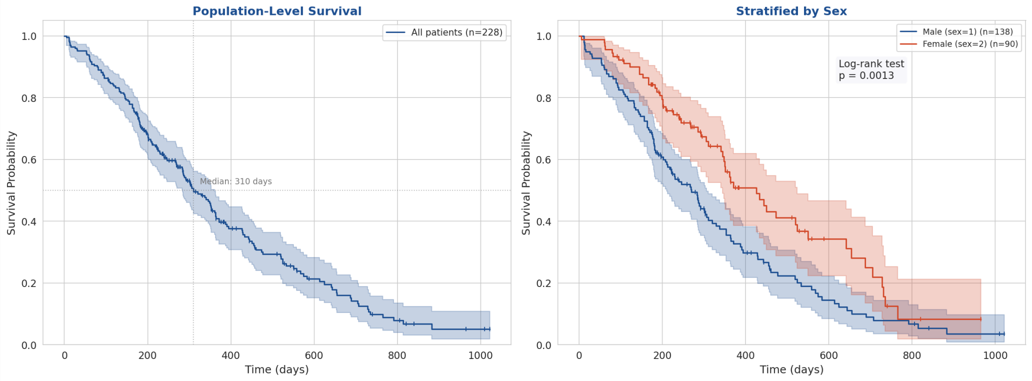
Kaplan-Meier: A Worked Example

Five patients from the censoring diagram. $\hat{S}$ starts at 1 and updates at each event:

Time	What happened	d_j	n_j	$1 - d_j/n_j$	$\hat{S}$ after
0–1.8 yr	(no events yet)				1.000
1.8 yr	Event (C)	1	5	4/5	$1 \times \frac{4}{5} = 0.800$
2.5 yr	Event (E)	1	4	3/4	$0.8 \times \frac{3}{4} = 0.600$
3.5 yr	Event (A)	1	3	2/3	$0.6 \times \frac{2}{3} = 0.400$
4.2 yr	Left study (D)	—	n drops from 2 to 1; $\hat{S}$ unchanged		
5.0+ yr	Still followed (B)	—	$n = 1$; no more events observed		

Between events, $\hat{S}$ is *constant*. E.g., $\hat{S}(2.0) = \hat{S}(2.4) = 0.800$.

K-M on Real Data: Population and Subgroups



NCCTG Lung Cancer dataset (lifelines). Left: population-level K-M curve with median survival. Right: stratified by sex—subgroup differences emerge. The point is how to *read* K-M curves: step functions, censoring marks (|), confidence bands, log-rank comparison.

Cox Proportional Hazards

Model the hazard as a function of covariates:

$$h(t \mid X) = h_0(t) \cdot \exp(\beta_1 X_1 + \cdots + \beta_p X_p)$$

- $h_0(t)$: baseline hazard (left unspecified—**semi-parametric**)
- $\exp(\beta_j)$: **hazard ratio** for covariate j. HR > 1: higher risk. HR < 1: protective.
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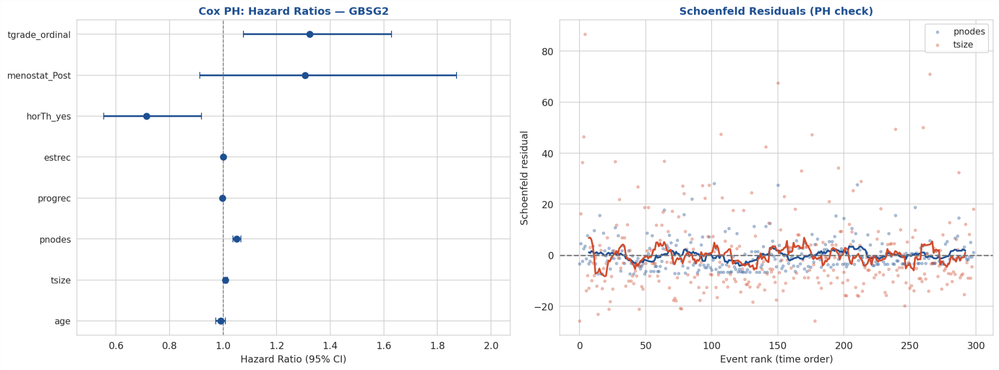
Are these hazard ratios causal? Only if confounders are controlled. From a registry: **associational**. From an RCT: **causal**. The data source determines the interpretation, not the model.

Cox PH on Real Data (Notebook Fig. 4)



GBSG2 breast cancer RCT (lifelines, 686 patients). Left: hazard ratios with 95% CI (features ordered by effect size). Right: **Schoenfeld residuals**—a diagnostic for the PH assumption. Each covariate gets a residual at each event time; if PH holds, these should show no trend. A slope means the covariate's effect *changes* over time (e.g., a drug that works initially but wears off). Flat-looking patterns are consistent with PH for that covariate.

Beyond Cox: Modern Survival Models

Cox PH assumes the log-hazard is a *linear* function of covariates. What if it isn't?

DeepSurv (Katzman et al., 2018): replace $\beta^\top X$ with a neural network $f_\theta(X)$, keep the same loss function (partial likelihood). Helps when there are non-linear interactions and large n.

Other approaches:

- **Random survival forests:** ensemble methods for survival data
- **Discrete-time models:** bin time, use standard classifiers per bin
- **DeepHit** (Lee et al., 2018): directly estimate the probability of events in each time bin

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Caveat: On small tabular datasets (like GBSG2), Cox often matches or beats neural approaches. Depth helps most with high-dimensional inputs (imaging, genomics) and large n. See notebook appendix for a DeepSurv vs. Cox comparison.

Synthesis

Back to Our Question

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3. **RCT:** Randomization balances confounders in expectation—but expensive, narrow population, possibly unethical for surgical comparisons.

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3. **RCT:** Randomization balances confounders in expectation—but expensive, narrow population, possibly unethical for surgical comparisons.
4. **In practice:** Combine evidence sources. Interpret results in light of the data source. Meta-analyses synthesize across studies.

Thursday (L24): Causal inference methods that let us extract stronger answers from observational data—and what happens when the *outcome you're measuring* is itself biased.

Summary

Part I: Population Health Data

- The question you're asking determines the data you need: descriptive, predictive, or causal.
- EHR data is convenient but confounded. Registries improve coverage but don't eliminate confounding (Simpson's Paradox). RCTs address confounding by randomization but are expensive and narrow.
- Biobanks blur the designed/byproduct boundary. Volunteer-based cohorts are especially vulnerable to healthy volunteer bias.
- Causal inference methods can strengthen observational analyses—and reveal when outcomes themselves are biased. Thursday's topic.

Part II: Survival Analysis

- Time-to-event outcomes require handling censoring and competing risks.
- K-M estimates population survival; Cox PH models hazard as a function of covariates.
- Whether hazard ratios are causal depends on the data source, not the model.

Questions?

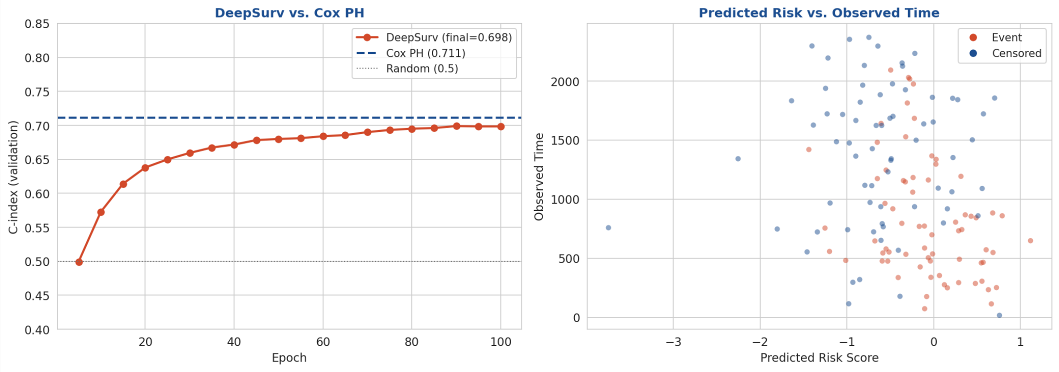
Appendix: How Cox Is Fit — The Partial Likelihood

Key insight: We never estimate $h_0(t)$. At each event, we ask: among those still at risk, did the right one die?

$$L(\beta) = \prod_{i:\, \delta_i = 1} \frac{\exp(\beta^\top X_i)}{\sum_{j \in \mathcal{R}(t_i)} \exp(\beta^\top X_j)}$$

Each term is a **softmax** over the risk set. Only ranking matters—the concordance index (C-index) is a standard evaluation metric. DeepSurv uses exactly this loss with $f_\theta(X)$ replacing $\beta^\top X$.

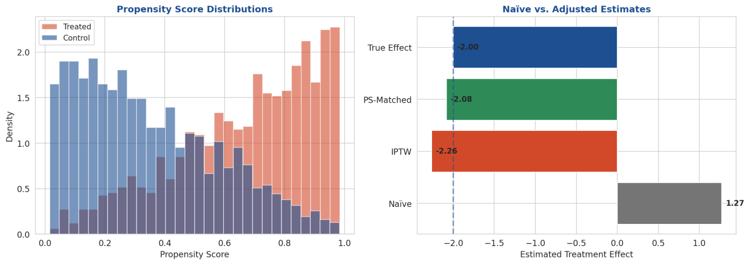
Appendix: DeepSurv vs. Cox (Notebook Fig. 5)



GBSG2: DeepSurv (C=0.698) slightly underperforms Cox (C=0.711). Expected on small tabular data with few covariates. Single-seed demo.

Appendix: Propensity Scores (Preview for L24)

Propensity score $e(X) = P(\text{Treatment} = 1 \mid X)$. Match or reweight patients to approximate randomization from observational data. Only adjusts for *measured* confounders.



Simulated data with known true effect. Naïve estimate is biased; PS adjustment recovers closer to truth. Full treatment Thursday (L24).

Appendix: References

Simpson's Paradox:

Charig et al. (1986). Renal calculi treatment. *BMJ*.

Registries & population data:

SEER: `seer.cancer.gov`

NHANES: `wwwn.cdc.gov/nchs/nhanes`

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Batty et al. (2020). Risk factor associations in UK Biobank vs. representative cohorts.

Murthy et al. (2004). Trial participants vs. cancer population.

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Survival analysis:

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Appendix: Trial Phases & Costs

Appendix: CDISC/SDTM Data Standards

CDISC standardizes trial data for FDA submission: Demographics (DM), Adverse Events (AE), Labs (LB), Vital Signs (VS). Each domain is a flat table with standardized variable names.

Analogy: DICOM for imaging (L21), ICD/LOINC for EHR codes (L17), CDISC for trials.